Critical Infrastructure Security

IY5612 (IY3612/IY4612) — Academic Year 2024/2025 (Unit 7)

Stephen D. Wolthusen
stephen.wolthusen@rhul.ac.uk

Department of Information Security and Communication Technology
Norwegian University of Science and Technology, Norway

and

Department of Information Security
Royal Holloway, University of London, United Kingdom

Unit 7— State Estimation

Contents

SCADA and DCS Systems

State Estimation

State Estimation
- Introduction
- Linear State Estimators
- Non-Linear State Estimators
- Attacks on State Estimators

Security of Control Systems Components

Part II

State Estimation

State Estimation

In this block the concept of state estimation is introduced:

- Basic concepts of state estimation
- Linear state estimators
- Non-linear state estimators
- Attacks on state estimators

Recommended Reading (1)

- Y. Liu, P. Ning, M. K. Reiter:
  False Data Injection Attacks against State Estimation in Electric Power Grids
  In: Proceedings of the 16th ACM Conference on Computer and Communications Security (ACM CCS 2009), Chicago, IL, USA, Nov. 2009, pp. 21–32
- K. Dehghanpour, Z. Wang, J. Wang, Y. Yuan, F. Bu:
  A Survey on State Estimation Techniques and Challenges in Smart Distribution Systems
  IEEE Transactions on Smart Grid Vol. 10, No. 2 (March 2019)

State Estimation

To be able to predict a (engineering, information, etc.) system's state, one would ideally like to have a fully observable system

- Recall the observability criterion — this implies controllability as well

It may not always be possible to satisfy this observability criterion, i.e. the complete state is not accessible or can otherwise not be measured

- This may be the case both for a static state, but particularly for any changes arising where we may not even "see" all of the changes

It may still be possible to give an **estimate** of the system's state if sufficient boundary conditions are known or assumed and state changes over time can be bounded

Application Domain Examples

State estimation problems arise in many situations and very different formulations:

- A network firewall or intrusion detection system must solve discrete state estimation problems as it must e.g. infer the state of a protocol session without necessarily having insight into either endpoint's state or observing all messages
- On start-up, a GPS navigation system will take time to acquire satellite signals; to be useful in this time or when the satellite signals become unavailable, current location has to be estimated (discretely or continuously)
- The state of power networks must typically be estimated and predicted to be able to adjust generation and/or load instantly
- Many cyber-physical systems from jet engines to manufacturing processes rely on **digital twins** — prime examples of state estimators with high stakes

State Estimation
Example Applications

State estimation is also relied upon in general advanced control systems, examples of which include:

Feedforward Control Control can anticipate estimated disturbances

Cascade Control Nested control loops can rely on estimated states where it is not possible to “build” states at each level

LQ Optimal Control Linear Quadratic optimal control seeks to base feedback on estimated states

MPC Model-Based Predictive Control seeks to predict future behaviour and optimises control based on estimated present state

State Estimation
Basic Properties

Any state estimate will contain errors:

- The estimate is, by definition, an approximation of the true state (which typically cannot be established)
- For physical systems, measurements will themselves contain errors from this notional true state owing to limited precision and accuracy
- Even for purely logical systems, we will often not be able to guarantee accuracy of all observed influences on state

State estimation seeks to minimise the remaining error by repeatedly updating the estimate based on knowledge of the system being observed and multiple measurements

- State estimation is hence only possible if we can obtain values for the estimation error

Linear State Estimators (1)

The simplest case for state estimation (both continuous and discrete) can be found for linear systems:

- The system's state can be described by linear equations
- For linear time-invariant (LTI) systems, the systm's output and future state will be determined only by current state and current inputs
- Even here the initial error will be relatively large, and several iterations are needed before the estimate becomes stable (converges)

For a state estimator to be useful e.g. for control, it must converge quickly, and algorithms must be sufficiently efficient

- An example for linear systems is hence a simple weighted least squares algorithm

Linear State Estimators (2)

(Linear) state estimators typically make a number of assumptions about the errors encountered:

- Measurements are – as always – noisy random variables
- Generally variables are assumed to be independent, but some are known to be correlated
- While noise can be correlated, it can generally be considered "smooth"

Non-Linear State Estimators

Not all “interesting” systems can be approximated successfully by linear models

- A number of approaches exist that depend on whether the underlying system can be described by non-linear but known functions or state models, and if a nominal solution even exists (e.g. Extended Kalman Filter)
- The main task remains to predict the propagation of state — this becomes more challenging if measurement errors and disturbances are not small
- Information systems are unfortunately often in this category: Discrete state systems whose transitions cannot be approximated
- In extreme cases explicit Bayesian models may be needed, but approaches such as Particle Filters (Sequential Monte Carlo methods) can be used to “extract” unknown (hidden) parameters from observations

Attacks on State Estimators

State estimators vary considerably in the size of the system they must analyse — some of the largest involve the electric power grid

- With smart grid developments operating the grid closer to its safe margins, the estimation has to be precise, timely, and needs to give immediate information for varying electricity generation or managing loads
- As physical sensors and interconnections can fail, state estimators are normally configured so they can deal with “bad” data
- Such bad data is conventionally identified by strong deviations from previous measurements, inconsistencies, or other simple statistical and probabilistic tests

Bad Data in State Estimators (1)

Tests for “ordinary” bad data — essentially a form of anomaly detection — can usually rely on some assumptions:

- Errors will normally be independent, and will not interact
- It is possible for real failures to cause multiple correlated measurements to be in error, and tests must account for these as well
- Measurements can also be checked for consistency if a sufficient degree of redundancy can be achieved

However, adversaries may also manipulate measurements maliciously for their own purposes:

- Forcing state estimator to a particular (erroneous) state

Bad Data in State Estimators (2)

In power systems, a bad state estimate can result in grid instability or outright failures:

- Operators are required to maintain frequency in a narrow band around 50Hz in the UK
- Deviations can cause fines to be levied, but in more extreme cases of mis-matches between supply and demand the grid becomes unbalanced and may need to take emergency measures up to local blackouts or generators tripping

Automation of power equipment makes recovery from such failures more challenging, particularly if physical changes or repairs are required after a failure

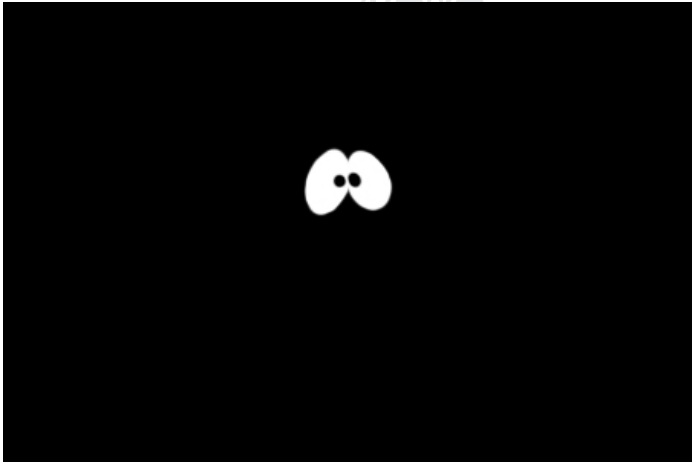
Bad Data in State Estimators (3)

State estimation provides a way of obtaining an approximation of observability for systems that cannot be fully observed

- The loss of estimated state is hence similarly problematic to a loss of view in a regular control system
- Equipment failure is normally — but not always — prevented by additional safeguard mechanisms, but these are also increasingly electronic
- Unlike e.g. in some industrial control system, there is no meaningful safe set-point that a power network can fall back on: Loss of view inevitably has negative consequences

Modern **digital twins** found in many areas from gas turbines to machine tools obviously rely on state estimation

Bad Data in State Estimators (3)



Bad Data in State Estimators (3)

To cause this equivalent to loss of view (estimate) it is sufficient to keep the state estimator from producing a result ("convergence") at the time it is needed

Careful analysis shows that different types of denial of service attacks are possible

- by suppressing measurements and their (timely) communication, respectively, or
- by feeding values causing the state estimation algorithm itself to fail — as opposed to driving the estimated state

Summary

This section provided material on

- Basic concepts of state estimation
- Linear state estimators
- Non-linear state estimators
- Attacks on state estimators